



Interfacing a Digital Camera
with a GRABLINK Board

Sony XCL-5005CR

Main characteristics.....	2
Configurations.....	3
Connection Information.....	4
Camera Setup.....	5

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Main characteristics

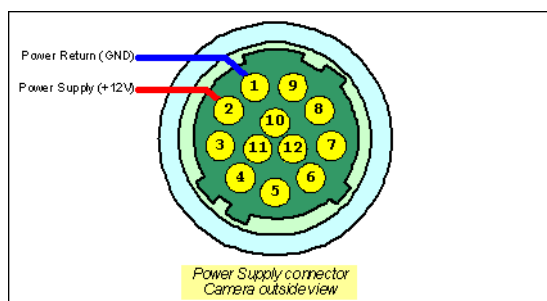
Sensor	Area-scan, color
Scanning method	PROGRESSIVE
Image size	2448 (H) x 2048 (V) Pixels
Frame rate	Up to 15 frames per second
Tap configuration	BASE_2T12
Last update	28 Nov 2011

Configurations

Configuration	CAM file	Description
P15RG	XCL-5005CR_P15RG.cam	Progressive Area-Scan, Asynchronous Reset, Grabber controls exposure.
P15RG_CFA	XCL-5005CR_P15RG_CFA.cam	Progressive Area-Scan, Asynchronous Reset, Grabber controls exposure, Color BAYER
P15SC	XCL-5005CR_P15SC.cam	Progressive Area-Scan, Free-Run Scanning.
P15SC_CFA	XCL-5005CR_P15SC_CFA.cam	Progressive Area-Scan, Free-Run Scanning, Color BAYER

Connection Information

Camera Power Supply Connector Pin-out



Valid for all configurations when the camera is supplied via the Power Supply connector

Camera Setup

Using the RS-232 Interface

The camera is configured via an ASCII terminal or from a PC running terminal emulator software. Below is the description of the ASCII codes to install the different configurations.

To configure the camera:

- Create the channel
- Restore the camera factory settings
- Configure the camera with the ASCII codes described below (carriage return ends each code)

Communication setting of Windows HyperTerminal

Baud Rate	38400bps
Data Length	8bit
Start Bit	1bit
Stop Bit	1bit
Parity	None
Xon/Xoff Control	None

Valid for all configurations

Command	Setting	Effect
PARTIAL	0	Partial Scan Setting: OFF
HPARTIAL	0	Horizontal Partial Scan Setting: OFF

Valid for configuration P15SC (RGB)

Command	Setting	Effect
IMG-TAP	0	Image Output Data Tap: 80 MHz 1TAP output
IMG-WIZE	3	Image Output Data Width: RGB 24 bit output
TRG-MODE	0	Trigger Mode Setting: Normal output of moving picture
SHUTTER	0-14	Shutter Speed Setting: as required by the application

Valid for configuration P15RG (RGB)

Command	Setting	Effect
IMG-TAP	0	Image Output Data Tap: 80 MHz 1TAP output
IMG-WIZE	3	Image Output Data Width: RGB 24 bit output
TRG-MODE	2	Trigger Mode Setting: External trigger/ Pulse width detection
TRG-POL	1	External Trigger Pulse Polarity Setting: Positive
EXTTRG	0	External Trigger Signal Input Selection: Camera Link

Valid for configuration P15SC_CFA

Command	Setting	Effect
IMG-TAP	1	Image Output Data Tap: 40 MHz 2TAP output
IMG-WIZE	0	Image Output Data Width: 12bit output
TRG-MODE	0	Trigger Mode Setting: Normal output of moving picture
SHUTTER	0-14	Shutter Speed Setting: as required by the application

Valid for configuration P15RG_CFA

Command	Setting	Effect
IMG-TAP	1	Image Output Data Tap: 40 MHz 2TAP output
IMG-WIZE	0	Image Output Data Width: 12bit output
TRG-MODE	2	Trigger Mode Setting: External trigger/ Pulse width detection
TRG-POL	1	External Trigger Pulse Polarity Setting: Positive
EXTTRG	0	External Trigger Signal Input Selection: Camera Link

Other parameters will be set as required by the application.